

HOTEL REVENUE DASHBOARD

T11 Mini Project – Business Recommendation & Reflection Report

Student: Chinta Karthikeya Santosh Reddy **Task:** T11 **PRN:** 26030242014

Theme: No-code Apps & Dashboards **Hotel:** Hotel Godavari Residency (fictional)

Dataset: 1,500 synthetic bookings for a 120-room hotel

1. Executive Summary

This project develops an AI-assisted, no-code hotel revenue dashboard from the supplied synthetic booking workbook. The dashboard focuses on five required management indicators—occupancy, ADR, RevPAR, cancellation rate and realised net revenue after channel commission—and extends the analysis into channel profitability and capacity-based overbooking strategy.

The verified dataset contains 1,500 bookings, of which 1,170 are confirmed and 330 are cancelled. The overall cancellation rate is 22.00%. Across the observed 414-day stay window, confirmed bookings account for 2,858 room nights. Using 120 rooms as capacity gives 49,680 available room nights, resulting in 5.75% occupancy. Confirmed gross room revenue is ■18,491,932 and realised net revenue after stated channel commissions is ■17,028,493.16.

Channel analysis shows meaningful differences in realised net revenue per booking. Walk-in, Corporate and Direct Website bookings record higher realised value per booking than the two OTA channels in this dataset, while the OTA channels also have higher recorded cancellation rates. Because the dataset is observational, these differences are treated as descriptive rather than causal.

The capacity test found a peak gross booking load of 22 rooms and a peak confirmed load of 19 rooms against 120 rooms of capacity. A conservative 90% review trigger equals 108 rooms, and the historical data never reached it. Because no-show behaviour and oversell costs are not present, the project does not claim an optimal overbooking quantity.

The resulting recommendation is to strengthen direct and corporate demand where feasible, retain OTA channels selectively for incremental demand, monitor realised value and cancellations together, and use 90% capacity as a review trigger rather than an automatic overbooking rule.

2. Business Problem and Analytical Approach

The business problem is to turn a booking-level dataset into a management dashboard that can answer three practical questions: How is the hotel performing? Which channels produce realised value? And does the historical data justify an overbooking strategy?

The supplied workbook contains booking date, check-in date, stay length, room type, channel, segment, guests, average daily rate, channel commission rate, cancellation status and guest rating. There is no direct revenue field, so revenue must be derived from nights and average daily rate. This distinction was important because the initial AI-assisted analysis could otherwise assume columns that do not exist.

Revenue calculation. For confirmed bookings, gross room revenue is calculated as nights × average daily rate. Net revenue is then calculated after the stated channel commission. Cancelled bookings contribute zero realised room revenue.

Occupancy calculation. Occupancy is confirmed room nights divided by available room nights. Available room nights are calculated as 120 rooms × 414 days, where the 414-day period is the observed analysis stay window from 03 Apr 2025 to 22 May 2026.

ADR and RevPAR. ADR is confirmed gross room revenue divided by confirmed room nights. RevPAR is confirmed gross room revenue divided by all available room nights. Net RevPAR is additionally shown as a supporting metric using realised net revenue.

3. Verified KPI Results

Metric	Result	Interpretation / basis
Total bookings	1,500	All booking records
Confirmed bookings	1,170	Cancelled = No
Cancelled bookings	330	Cancelled = Yes
Cancellation rate	22.00%	330 / 1,500
Confirmed room nights	2,858	Confirmed bookings only
Available room nights	49,680	120 × 414 days
Occupancy	5.75%	2,858 / 49,680
ADR	■6,470.24	Confirmed gross revenue / confirmed room nights
RevPAR	■372.22	Confirmed gross revenue / available room nights
Realised net revenue	■17,028,493.16	After stated channel commission
Net RevPAR	■342.76	Net revenue / available room nights

The low occupancy figure should be read in the context of the dataset's observed stay window. It is a mathematically consistent result under the stated denominator, not evidence by itself that the hotel should change its operating model.

4. Channel Profitability Analysis

The channel analysis compares realised net revenue per booking and places cancellation rate beside it. This avoids interpreting gross booking volume as equivalent to realised revenue.

Channel	Bookings	Cancelled	Cancel rate	Net / booking
Walk-in	80	15	18.75%	■14,315
Corporate	308	52	16.88%	■13,829
Direct Website	314	54	17.20%	■12,689
Travel Agent	139	19	13.67%	■12,360
OTA - MakeMyTrip	441	125	28.34%	■9,212
OTA - Booking.com	218	65	29.82%	■8,529

Walk-in records ■14,315 in realised net revenue per booking, followed by Corporate at ■13,829 and Direct Website at ■12,689. Travel Agent records approximately ■12,360. The two OTA channels record ■9,212 for MakeMyTrip and ■8,529 for Booking.com. MakeMyTrip also has the highest booking volume at 441 records, which shows why volume and value should be monitored together.

Cancellation rates vary across channels. Booking.com records 29.82%, MakeMyTrip 28.34%, Walk-in 18.75%, Direct Website 17.20%, Corporate 16.88% and Travel Agent 13.67%. These figures are historical observations from this dataset. They do not demonstrate that the channel itself causes a cancellation.

5. Capacity and Overbooking Strategy Test

A conservative capacity-triggered strategy was proposed because the dataset does not contain no-show behaviour. The proposed review point is 90% of the 120-room capacity, or 108 rooms. The purpose of this threshold is to prompt a management review when demand becomes materially closer to capacity; it is not an assertion that the hotel can safely sell 108 plus a fixed number of additional rooms.

The historical test calculated both gross booking load and confirmed room load by date. The highest gross booking load was 22 rooms, while the highest confirmed occupied load was 19 rooms. Therefore, the 108-room review trigger was never reached in the supplied historical data.

The absence of no-show information is a material limitation. Cancellation and no-show are different events, so the 22% cancellation rate cannot simply be converted into an overbooking buffer. Before operational activation, management would need measured no-show behaviour, same-day cancellation behaviour, expected replacement demand and the financial/operational cost of relocating an oversold guest.

6. Next-Season Revenue Strategy

Action 1 — Strengthen direct and corporate demand. The historical results show higher realised net revenue per booking for Corporate and Direct Website than the two OTA channels. The hotel can test direct-booking offers, corporate relationships and repeat-customer incentives, while measuring whether the additional demand remains profitable.

Action 2 — Continue using OTA channels selectively. The evidence does not support eliminating OTA channels. They contribute meaningful volume, particularly MakeMyTrip. Instead, management should monitor OTA volume, cancellation rate and realised net revenue together and evaluate whether incremental OTA demand fills otherwise unused capacity.

Action 3 — Introduce a capacity review trigger. When daily committed/confirmed demand approaches 108 rooms, management should review channel mix, recent cancellation/no-show behaviour and expected remaining demand. Overbooking should be considered only after those additional operational measures are available.

Action 4 — Track revenue quality, not just bookings. Weekly or periodic management review should bring together occupancy, ADR, RevPAR, cancellation rate and realised net revenue per booking. A high booking count can coexist with cancellations and channel commission costs, so bookings alone are not a complete revenue measure.

7. Dashboard Design and Business Use

The refined dashboard is organised around an executive-first layout. The five required KPIs appear at the top so that a manager can immediately read the overall position. The channel section then explains how realised value differs across acquisition sources. The capacity section separates historical demand from the proposed decision trigger, reducing the risk of interpreting the strategy as a blanket overbooking recommendation.

The dashboard also includes an executive takeaway area that translates the calculations into three management observations: revenue quality, cancellation exposure and capacity pressure. The supporting Excel workbook provides the underlying analytical sheets so that the headline values can be reconciled to the source data.

8. AI-Assisted Development and Iteration

The project was intentionally developed iteratively rather than as a single prompt. The process began with dataset inspection, where the AI was instructed not to assume missing columns. KPI formulas were then defined, the actual stay dates were checked, and the baseline calculations were independently verified. Subsequent iterations focused on dashboard structure, channel profitability, a conservative overbooking proposal, historical capacity testing, stress-testing, next-season strategy and final wording.

The AI Use Log contains 15 documented iterations. It records what each prompt was intended to accomplish and how the output was accepted, revised or constrained. This provides an auditable record of the AI-assisted development process rather than presenting the final dashboard as a one-shot generation.

9. Responsible AI, Limitations and Reflection

The supplied data is synthetic and created for teaching. No real guest personal or confidential information was used. AI-generated calculations were not accepted without verification: the headline figures were recomputed against the supplied booking records before being used in the final dashboard and report.

The main analytical limitation is the lack of no-show behaviour. This prevents the project from estimating a statistically or financially optimal overbooking quantity. The strategy therefore remains a decision framework that can be tested once the hotel has additional operational data.

A second limitation is the observed stay-window denominator. The workbook provides the booking period and the actual check-in dates, but it does not establish continuous hotel operations outside the observed stay period. For that reason, the final dashboard labels the 03 Apr 2025–22 May 2026 period as an 'analysis stay window' rather than claiming that the hotel operated continuously for 414 days.

A third limitation is that channel comparisons are descriptive. Higher or lower realised value associated with a channel does not establish causation because other factors—such as booking characteristics, room type, segment and timing—may differ between channels.

Reflection. The project demonstrated that AI is most useful when treated as an iterative development partner rather than an unquestioned source of answers. The most important changes came from verification: checking the actual workbook columns, correcting the treatment of cancelled bookings, validating the analysis period, testing capacity against actual daily loads and explicitly separating cancellation from no-show behaviour. The final output is therefore more conservative than a generic AI-generated hotel strategy, but it is more closely tied to what the supplied data can actually support.

10. Final Submission Contents

The final project package contains the refined HTML dashboard, the refined Excel workbook with analytical output sheets, the standalone AI Use Log, and this expanded report. The GitHub repository contains these files in clearly labelled folders so that the dashboard, evidence and supporting analysis can be accessed directly.

11. Final Verification Statement

The final figures used throughout the submission are: 1,500 bookings; 1,170 confirmed bookings; 330 cancellations; 22.00% cancellation rate; 2,858 confirmed room nights; 49,680 available room nights; 5.75% occupancy; ■6,470.24 ADR; ■372.22 RevPAR; ■17,028,493.16 realised net revenue; ■342.76 net RevPAR; 22-room peak gross booking load; and 19-room peak confirmed load.

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